

Task-4 Block Diagram

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ARM Cortex-A55 Cluster (linux)

- YOLO26n pre/post processing (resize, normalize, NMS)
- Voice synthesis (pytt3x3, espeak)
- B96 LTE comms, GPS, escalation logic
- Top level app orchestration

memory bus
(frame in,
detections out)

interrupt
only (confirm
fall)

Hexagon DSP / NPU

- Runs YOLO26n inference
- Target: 4x FPS @ 0.3x power vs CPU-only (vendor claim)
- Bandwidth bound past that point (Section-A)

Cortex-M7 coprocessor (bare metal firmware)

- Fall detection FSM, 5 states, 178 cycles/sample
- 100 MHz timing closed with margin
- Asleep, 99.98% time (clock-gated)

memory bus

PMA,
no polling

Sensors (SPI/I2C, 100Hz)

- ICM-42688-P (IMU)
- VL53L5CX (ToF)
- connected to M7 only.

Camera Flow: Return path loop

